

- The indicator variables for rank have a slightly different interpretation. For example, having attended an undergraduate institution with rank of 2, versus an institution with a rank of 1, changes the log odds of admission by -0.675.

You can also exponentiate the coefficients and interpret them as odds-ratios. Now we can say that for a one unit increase in gpa, the odds of being admitted to graduate school (versus not being admitted) increase by a factor of 2.23.

```
library(dplyr)
model %>%
  broom::tidy(exp = TRUE)
```

We can look at the overall quality of the model with the glance() function

```
library(dplyr)
model %>%
  broom::glance()
```

Poisson Regression

We assume that the response variable y_i is a count, such that the observation $Y_i = 0, 1, 2, \dots$. A reasonable probability model for count data is often the Poisson distribution.

$$f(y) = \frac{e^{-\mu} \mu^y}{y!}, y = 0, 1, \dots$$

where the parameter $\mu > 0$.

For the Poisson distribution

$$E(Y) = \mu = Var(Y)$$

The Poisson regression model can be written as:

$$Y_i = E(Y_i) + \varepsilon_i, i = 1, 2, \dots, n$$

We assume that the expected value of the observed response can be written as:

$$E(y_i) = \mu_i$$

and there is a function g that relates the mean response to a linear predictor say

$$g(\mu_i) = \eta_i = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k = \mathbf{x}_i' \boldsymbol{\beta}$$

The function g is usually called the **link function**.

The relationship between the mean and the linear predictor is

$$\mu_i = g^{-1}(\eta_i) = g^{-1}(x_i' \boldsymbol{\beta})$$

The link function for the Poisson distribution is the **log link**

$$g(\mu_i) = \ln(\mu_i) = x_i' \beta$$

The relationship between the mean of the response variable and the linear predictor is

$$\mu_i = g^{-1}(x_i') = e^{x_i' \beta}$$

The log link is particularly attractive for Poisson regression because it ensures that all of the predicted values of the response variable will be nonnegative.

The method of maximum likelihood is used to estimate the parameters in Poisson regression.

The likelihood function is:

$$\begin{aligned} L(y, \beta) &= \prod_{i=1}^n f_i y_i = \prod_{i=1}^n \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \\ &= \frac{\prod_{i=1}^n \mu_i^{y_i} \exp\left(-\sum_{i=1}^n \mu_i\right)}{\prod_{i=1}^n y_i!} \end{aligned} \quad (32)$$

Where $\mu_i = g^{-1}(x_i' \beta)$. Once the link function is selected we maximize the log-likelihood

$$\ln L(y, \beta) = \sum_{i=1}^n y_i \ln(\mu_i) - \sum_{i=1}^n \mu_i - \sum_{i=1}^n \ln(y_i!)$$

Iteratively reweighted least squares can be used to find the maximum-likelihood estimates of the parameters in Poisson regression, following an approach similar to that used for logistic regression.

Once the parameter estimates $\hat{\beta}$ are obtained the fitted Poisson Regression model is

$$\hat{y}_i = g^{-1}(x_i' \hat{\beta})$$

Example

The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational, general or academic) and the score on their final exam in math.

In this example, `num_awards` is the outcome variable and indicates the number of awards earned by students at a high school in a year, `math` is a continuous predictor variable and represents students' scores on their math final exam, and `prog` is a categorical predictor variable with three levels indicating the type of program in which the students were enrolled. It is coded as 1 = "General", 2 = "Academic" and 3 = "Vocational". Let's start with loading the data and looking at some descriptive statistics.

```
p <- read.csv("https://stats.idre.ucla.edu/stat/data/poisson_sim.csv")
p <- within(p, {
  prog <- factor(prog, levels=1:3, labels=c("General", "Academic",
                                           "Vocational"))
})
```

```
id <- factor(id)
})
```

At this point, we are ready to perform our Poisson model analysis using the glm function. We fit the model and store it in the object m1 and get a summary of the model at the same time.

```
summary(m1 <- glm(num_awards ~ prog + math, family="poisson", data=p))
```

Next come the Poisson regression coefficients for each of the variables along with the standard errors, z-scores, p-values and 95% confidence intervals for the coefficients. The coefficient for math is .07. This means that the expected log count for a one-unit increase in math is .07. The indicator variable progAcademic compares between prog = "Academic" and prog = "General", the expected log count for prog = "Academic" increases by about 1.1. The indicator variable prog.Vocational is the expected difference in log count ((approx .37)) between prog = "Vocational" and the reference group (prog = "General").